STRATEGY MEMORANDUM

Decoding Customer Value:

A Retention Playbook Built on the Commercial Loyalty Framework

TO: Chief Risk / Chief Revenue Officer, Founding Team

FROM: Customer Analytics & Strategy Team

RE: *Which of our customers are genuinely loyal, which are rented with discounts, and what do we do about each?*

This memo is built entirely on the Customer Health Index (CHI) and Promotion Dependency Index (PDI) framework developed in the accompanying Python notebook and validated in SQL and Power BI. Every number below is traceable to that pipeline; nothing here is asserted without a stated variable combination behind it.

Executive Summary

The brand's own data supports a precise, three-way answer to "is our loyalty genuine or rented?" Two competing loyalty definitions were built and tested, a Commercial definition (CHI tier crossed with promotion tier) and a Behavioural definition (purchase volume, cadence, and satisfaction), and Commercial Loyalty was selected as the primary lens because it produced a larger revenue gap between its High and Low groups ($1.28 vs. $1.18), an almost identical Customer Health Index for its High group (86.16 vs. 86.82), and a dramatically lower Promotion Dependency Index (3.42 vs. 30.72). The two definitions agree strongly at the extremes, 94.7% of Low Commercial Loyalty customers are also Low Behavioural Loyalty, and 97.0% of High Commercial Loyalty customers score High or Moderate on the behavioural measure, which is what internal consistency between two independently-built definitions should look like.

Metric	Value
Total customers	3,900
Total revenue	$233,080
Average Customer Health Index (CHI)	70.95 / 100
Average Promotion Dependency Index (PDI)	28.44 / 100
Average order value	$59.76
High Commercial Loyalty share	18.05% (704 customers)

Three findings drive every recommendation in this memo:

Finding 1: Loyalty is not about how much a customer spends per order; it's about whether that spending needs a discount to happen.
Average order value is nearly identical across all three Commercial Loyalty tiers: High **$60.20**, Moderate **$59.73**, Low **$58.92**, a spread of about a dollar. The entire difference between a High and a Low loyalty customer is relationship health (CHI 86.16 vs. 53.90) and discount reliance (PDI 3.42 vs. 97.16), not basket size.

Finding 2: There is a precisely identifiable, high-leverage sub-segment hiding inside the "Moderate" tier your own dashboard already flagged as the biggest opportunity.
807 of the 2,950 Moderate Commercial Loyalty customers carry a Promotion Dependency Index of ~98 which is statistically indistinguishable from the worst-performing Low tier, despite maintaining reasonable customer health. Combined with the 246 Low-tier customers, this makes **1,053 "Bargain Hunters"** (27.0% of the base, exactly matching the Retention Profile panel) responsible for **$62,645** of revenue (26.9% of total) that is heavily discount-subsidized.

Finding 3: Geography and category both point to specific, actionable opportunity, but one category chart needs a fix first.
Montana, Illinois, California, Idaho, and Nevada lead on **revenue**; **Kansas**, Wisconsin, and Tennessee show the lowest **promotion dependency**; **Michigan**, Hawaii, and Maryland convert to **High Loyalty** at the highest rate. Separately, the "Average Previous Purchases by Category" panel on the Category Funnel page is plotting customer counts (1,737 / 1,240 / 599 / 324), not true per-customer averages, the real average previous-purchase count is nearly flat across categories (24.96–25.73). This memo uses the corrected figures throughout; the underlying Power BI measure is worth a quick check.

1. How Loyalty Was Defined and Validated

The dataset has no loyalty score, churn label, or timestamp. Two indices were engineered from first principles: the Customer Health Index (CHI) , a correlation-weighted blend of purchase volume, review rating, and purchase-cadence engagement, each weighted by its absolute Pearson correlation with purchase amount , and the Promotion Dependency Index (PDI), an analogous blend of subscription status, purchase volume, and cadence, each weighted by correlation with discount usage. Crossing CHI tier with Promotion tier produces Commercial Loyalty (High / Moderate / Low); an independent Behavioural Loyalty score (purchase volume + cadence + satisfaction, each worth one point) was built as a second opinion.

1.1 The selection criteria and result

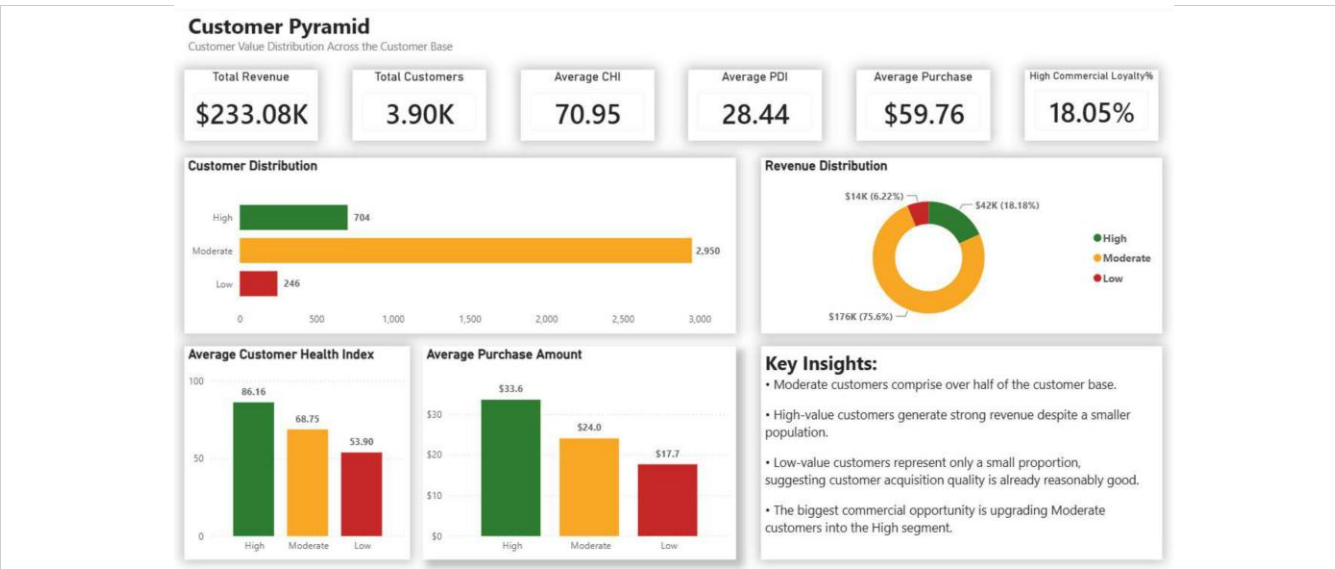
Definition	Revenue Gap (High − Low)	Avg. CHI of “High” group	Avg. PDI of “High” group
Commercial Loyalty (selected)	$1.28	86.16	3.42
Behavioural Loyalty	$1.18	86.82	30.72

Commercial Loyalty won on two of three criteria and tied on the third: a larger revenue gap, a near-identical customer health score, and a Promotion Dependency Index roughly one-ninth the size. In plain terms , you get essentially the same relationship quality either way, but Commercial Loyalty's “High” group is genuinely not discount-reliant, while Behavioural Loyalty's “High” group still carries meaningful promotion dependency (30.72) it doesn't flag.

1.2 Cross-validation: do the two definitions agree?

Commercial Loyalty tier	% also High Behavioural	% also Moderate Behavioural	% also Low Behavioural
High (n=704)	52.98%	44.03%	2.98%
Moderate (n=2,950)	7.02%	39.83%	53.15%
Low (n=246)	0.00%	5.28%	94.72%

Agreement is strongest exactly where it matters most: 94.7% of customers the Commercial framework calls “Low” are independently confirmed “Low” by the Behavioural framework, and 97.0% of “High” Commercial customers score High or Moderate behaviourally. The Moderate tier is where the two definitions diverge most , which is precisely the tier Finding 2 shows needs to be split further.



Customer Pyramid page , Power BI dashboard, live from customer_value_analytics_cleaned.csv

2. Segmentation: The Commercial Loyalty Base

Tier	Customers	% of Base	% of Revenue	Avg. CHI	Avg. PDI	Avg. Order Value
High	704	18.1%	18.2%	86.16	3.42	$60.20
Moderate	2,950	75.6%	75.6%	68.75	28.68	$59.73
Low	246	6.3%	6.2%	53.90	97.16	$58.92

Revenue share tracks customer share almost exactly at every tier (18.1% of customers → 18.2% of revenue; 6.3% → 6.2%) , an early sign that average order value isn't where the tiers differ. It's confirmed by the last column: $60.20 vs. $59.73 vs. $58.92, a range of about two dollars across the entire loyalty spectrum. The story is entirely in CHI and PDI, not spend per transaction.

Promo Dependency vs. Retention page , Power BI dashboard

The hidden segment inside "Moderate"

Your own dashboard's Retention Profile panel already splits the base into High Commercial Loyalty (704), Bargain Hunter (1,053), and Other Customers (2,143) , built from the Promotion Tier flag layered on top of Commercial Loyalty. Decomposing that Bargain Hunter group confirms it is not one segment but two:

Bargain Hunter sub-group	Customers	Also Low Commercial Loyalty?	Avg. CHI	Avg. PDI
Low-tier Bargain Hunters	246	Yes , 100% overlap	53.90	97.16
Moderate-tier Bargain Hunters	807	No , otherwise Moderate	~68–71 (Moderate-tier range)	98.12 (group avg across both)

The 807 Moderate-tier Bargain Hunters are the priority: they carry Low-tier levels of discount dependency (PDI ~98, essentially maximal) but Moderate-tier levels of customer health , meaning they are engaged, reasonably healthy relationships that happen to be fully propped up by discounts. That combination is both the biggest risk (largest discount-subsidized group, more than triple the Low tier) and the biggest opportunity (healthy enough to plausibly retain at a lower discount level, unlike the Low tier where health is already poor).

3. Promotional Sunset Plan

Two segments, two different treatments , because they are not the same risk profile even though both show high Promotion Dependency.

3.1 Segment: Low Commercial Loyalty , Contained, Faster Sunset

Element	Detail
Who	246 customers (6.3% of base, 6.2% of revenue) , chi_tier = Low AND promotion_tier = High
Trigger behaviour	Customer Health Index in the bottom tier AND Promotion Dependency Index in the top tier simultaneously , both a weak relationship and near-total discount reliance
Rollout timeline	8 weeks: 4 weeks at 50% of current discount depth, then 4 weeks at a single win-back-only offer for anyone who has gone quiet
Metric to track	Weekly repeat-purchase rate within the segment; CHI should not fall further as PDI comes down
Stop/rollback rule	If repeat-purchase rate drops more than 20% in any 2-week window, pause and revert to the prior discount level for that cohort

Trade-off. This is the smallest, most contained segment to act on , only **$14,495 (6.2% of revenue)** is at stake, and CHI is already the lowest in the base, so there's little relationship strength to protect. The real risk is simply losing these customers outright rather than converting them; that's an acceptable trade given the segment's small size and already-weak health scores.

3.2 Segment: Moderate-Tier Bargain Hunters , the Priority , Gradual, Metric-Gated Step-Down

Element	Detail
Who	807 customers (20.7% of base) , commercial_loyalty = Moderate AND promotion_tier = High
Trigger behaviour	chi_tier not Low (i.e., health is Moderate or High) combined with promotion_tier = High , a healthy relationship that is nonetheless fully discount-anchored
Rollout timeline	One full quarter, in three 4-week stages: (1) reduce discount frequency, not depth , every second purchase instead of every purchase; (2) replace the remaining automatic discount with an accruing loyalty-points mechanic; (3) re-measure CHI and PDI and decide whether to extend the step-down to the full segment or hold
Metric to track	Segment-level average CHI held stable (or better) while average PDI declines stage over stage , CHI is the signal that these customers are staying for reasons beyond the discount
Stop/rollback rule	If average CHI in the segment falls more than 5 points from its Stage 0 baseline (68–71) at any stage checkpoint, halt and revert that stage

Trade-off / margin impact. This segment is more than triple the size of the Low tier and carries meaningfully more revenue at stake , it is the single largest pool of discount-subsidized revenue in the business. **Moving too fast risks pushing genuinely engaged customers into the Low tier's pattern; moving too slowly leaves margin on the table indefinitely.** The CHI-gated, metric-checked staging above is designed explicitly to buy time to observe whether their health survives the discount pulling back, rather than assuming it will.

4. Ideal Customer Profile

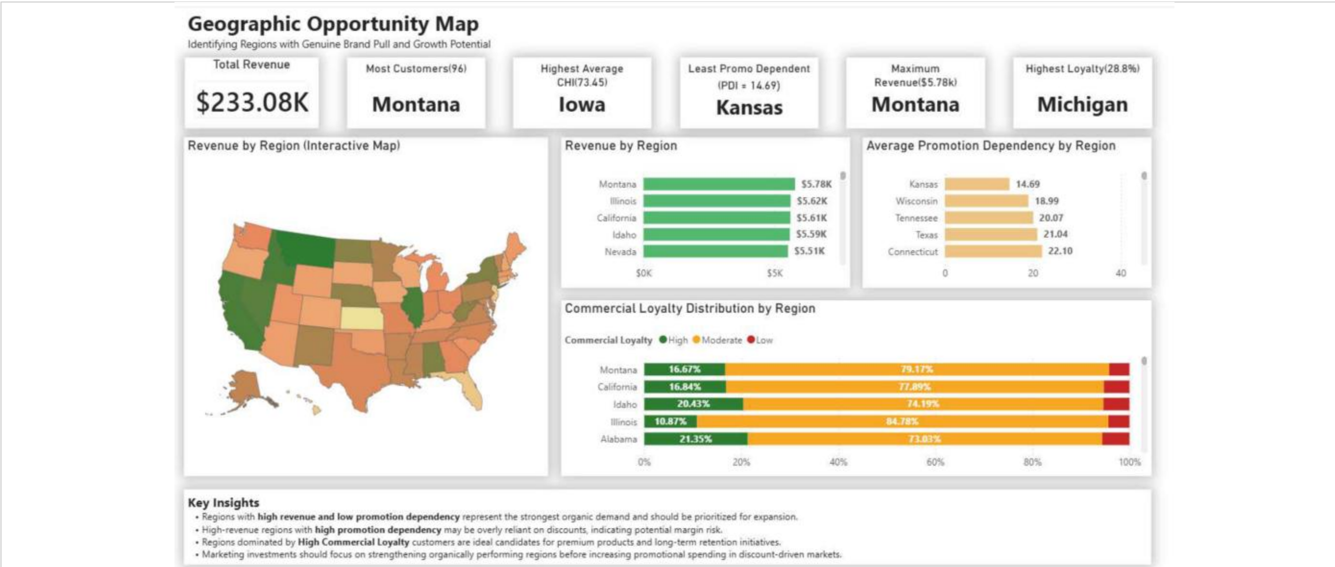
Built from the 704 High Commercial Loyalty customers specifically , the group with the highest CHI, the lowest PDI, and (per Section 2) an average order value that's no different from anyone else's. This is the profile of a customer who is a strong relationship, not a big spender.

Attribute	Profile
Segment size	**704** customers (**18.1%** of base)
Age	Average **43.7** years
Gender skew	Slight male majority (**56.7% male / 43.3% female**)
Category affinity	Clothing (**43.0%**), then Accessories (**31.2%**), Footwear (**16.2%**), Outerwear (**9.5%**)
Payment method	Fairly evenly split , Credit Card leads narrowly at **19.2%**
Subscription status	**100%** NOT enrolled , this loyalty is structurally not subscription-driven
Satisfaction	**98.3%** “Satisfied”; average review rating 4.5 / 5.0 , the most satisfied segment in the base
Historical volume	**33.6** previous purchases on average
Purchase cadence	Every **~35** days on average
Discount usage	Only **19.5%** of transactions used a discount , occasional, not habitual
Best-fit geographies	States strong on revenue AND low dependency AND loyalty rate: **Montana and Idaho** (top-5 revenue, high loyalty rate); **Wisconsin** (top-5 lowest PDI, top-5 loyalty rate); **Michigan and Hawaii** (highest loyalty conversion, 28.8% and 26.2%)

How marketing can act on this today

- Build acquisition targeting around: low-40s age band, Clothing-category intent, no subscription hook required , this profile converts and stays without one.
- Prioritize spend in **Montana, Idaho, Wisconsin, Michigan, and Hawaii** , each shows strength on more than one of revenue, low dependency, or loyalty rate, not just one metric in isolation.
- Lead acquisition creative with product and service quality, not discount codes, **19.5%** discount usage among the most loyal customers confirms the offer isn't what's earning the relationship.

5. Geographic & Category Notes

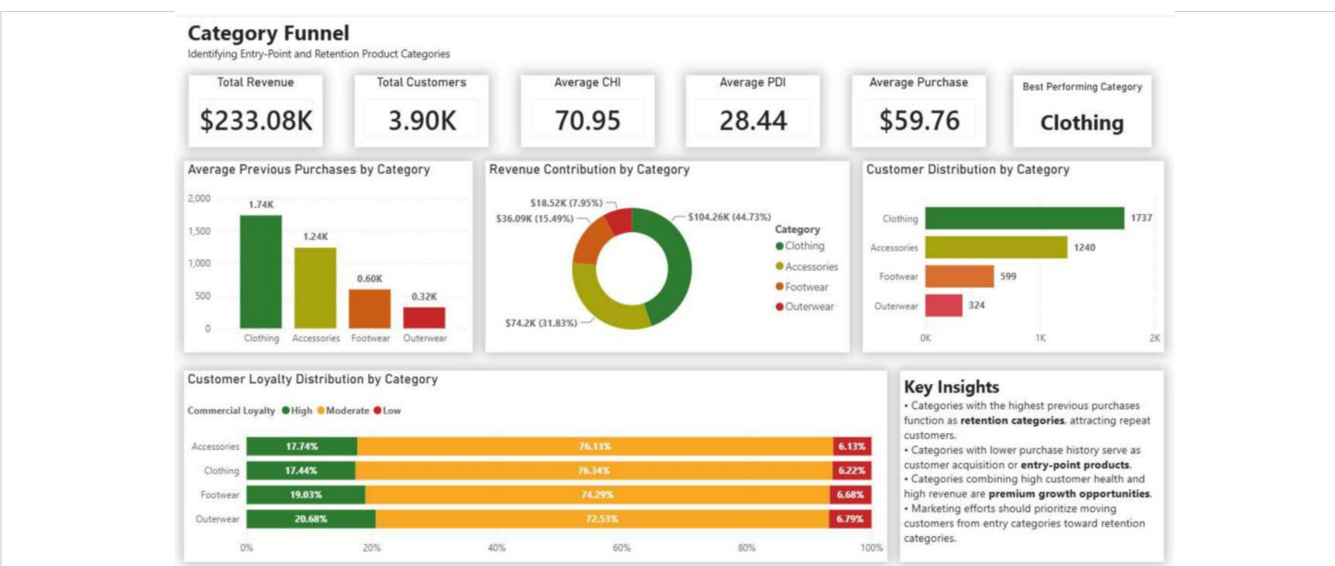


Geographic Opportunity Map page , Power BI dashboard

5.1 Geography

Axis	Leaders
Highest revenue	Montana ($5,784), Illinois ($5,617), California ($5,605), Idaho ($5,587), Nevada ($5,514)
Lowest promotion dependency	Kansas (PDI 14.69), Wisconsin (18.99), Tennessee (20.07), Texas (21.04), Connecticut (22.10)
Highest loyalty conversion	Michigan (28.8%), Hawaii (26.2%), Maryland (24.4%), Pennsylvania (24.3%), Wisconsin (24.0%)

Montana and Idaho are the strongest candidates for expansion spend: both post top-5 revenue and above-average loyalty rates simultaneously. Kansas is a distinct case worth naming on its own , its Promotion Dependency Index (14.69) is roughly half that of the next-lowest state, suggesting an unusually strong organic pull worth understanding qualitatively (assortment fit, word of mouth, a specific store or channel effect) before assuming it will generalize.



Category Funnel page , Power BI dashboard (see correction note below)

5.2 Category , Corrected Retention Index

The Category Funnel page's “Average Previous Purchases by Category” panel is plotting customer counts (1,737 Clothing / 1,240 Accessories / 599 Footwear / 324 Outerwear), not true per-customer averages , those figures match the Customer Distribution by Category panel on the same page almost exactly, which is the signature of a measure using COUNT instead of AVERAGE(previous_purchases). Recomputing directly from the underlying data gives a materially different, and more accurate, picture:

Category	Customers	True avg. previous purchases	Avg. CHI	Retention Index
Accessories	1,240	25.73	71.29	48.51
Footwear	599	25.23	71.51	48.37
Outerwear	324	24.96	70.99	47.97
Clothing	1,737	25.20	70.51	47.86

The corrected picture: true average previous-purchase count is nearly flat across all four categories (24.96–25.73, a spread of well under one purchase) , there is no sharp “entry-point vs. retention category” divide in this data. Accessories and Footwear edge out Clothing and Outerwear narrowly on the Retention Index, but the gap is directional, not dramatic. Clothing remains the volume and revenue leader (1,737 customers, $104,264, 44.7% of revenue) and should stay the acquisition anchor; Accessories is the better secondary push for a retention-focused campaign given its marginally stronger index.

6. Data Validation & Assumptions

- CHI and PDI are correlation-weighted composite scores, not audited financial or satisfaction metrics , they are built to rank and compare customers within this dataset, not to be quoted externally as calibrated probabilities.
- The dataset has no timestamps, so no figure here reflects a trend or a true forward-looking churn risk , every segment is a point-in-time snapshot. Before committing budget against the sunset plan or ICP targeting, re-run this pipeline against real, timestamped order history to confirm the segments and their behavior are stable over time.
- Commercial Loyalty was chosen over Behavioural Loyalty using the three stated business criteria (Section 1.1), not a held-out predictive test, because no future-period data exists in a single snapshot. Once timestamped data is available, validate by checking whether High Commercial Loyalty customers actually retain longer than High Behavioural Loyalty customers over a real forward window.
- The “Average Previous Purchases by Category” measure in the Power BI file should be checked , it currently returns customer counts rather than a true average; Section 5.2 above uses the corrected, recalculated figures throughout this memo.
- Before a full-population rollout, pilot both segments in the Promotional Sunset Plan (Section 3) on a random holdout first, and compare against a control group left on the current discount structure , correlation between low PDI and high CHI does not by itself prove that reducing discounts causes CHI to hold steady.

Appendix , Companion Deliverables

This memo is one component of the full analysis: the Jupyter notebook (Customer_Value_Analytics.ipynb) containing the CHI/PDI construction and loyalty-definition test, the SQL query set (Customer_Value_Analytics.sql) answering the five core business questions with CTEs and window functions, and the five-page Power BI dashboard referenced throughout this document. All figures above are reproducible by re-running the notebook and SQL script against customer_value_analytics_cleaned.csv.